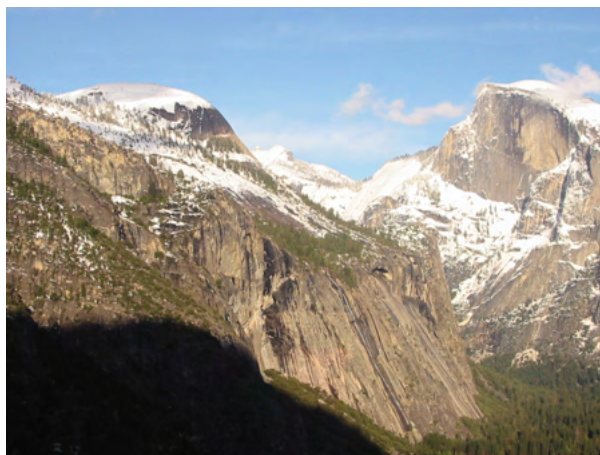


Assignment 1: Panorama Stitching

Step 1: Feature detection and description

Feature detection is performed on each individual image in the input. When taking photographs for a panorama, each successive photo will undergo a horizontal and vertical translation, a rotation, and variations in scale as well as differences in brightness and exposure. The feature detector will have to be invariant to photometric transformation and equivariant to geometric transformations in order to detect the same features at different points and perspectives in two different images. The SIFT detector was chosen as the feature detector, as it allows for image matching even when there are significant transformations or illumination differences between images.



(a) Image input for feature detection



(b) Image output with features labeled

Figure 1: Feature detection and keypoint extraction results.

Step 2: Feature matching and image alignment

After detecting the features for two images with overlap, we match the keypoints to determine the correspondence between descriptors in the two views. Feature matching between images allows for alignment of the detected features, as the movement of the camera will cause features to be at different positions in consecutive pairs of images. As well, differences in lighting, moving subjects, or repeating objects can result in outliers where features do not match between images. Brute force matching is used in the examples to find all matches between features in pairs of images. Although this method of matching will result in a higher proportion of bad matches than other matching algorithms, RANSAC is used here to discard these bad matches by selecting a match at random and counting all inliers (good matches) that are below a chosen maximum allowable residual error. If the number of inliers meets a chosen threshold, the matches that were calculated are acceptable.



(a) First input image for matching



(b) Second input image for matching

Figure 2: Input images for matching.

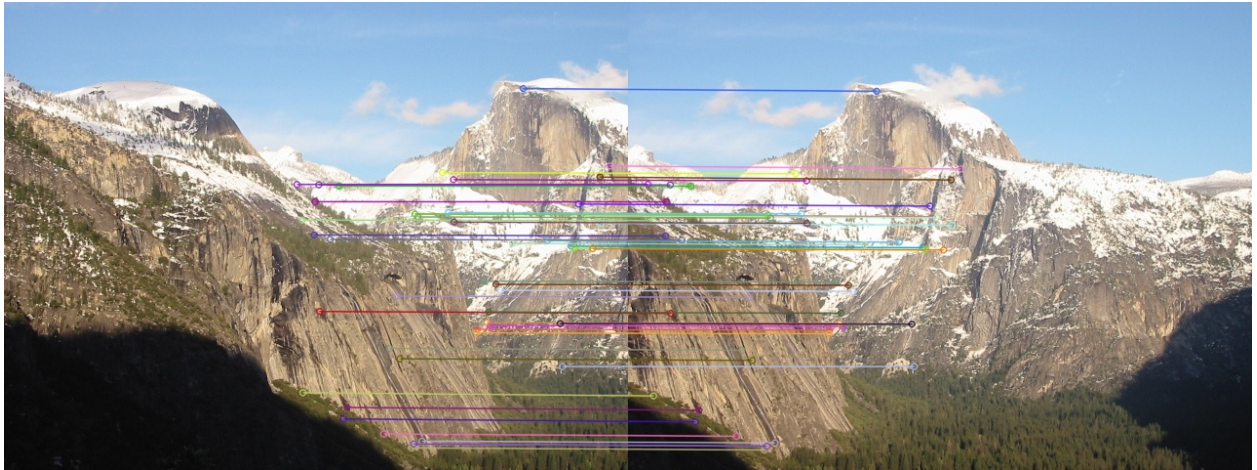
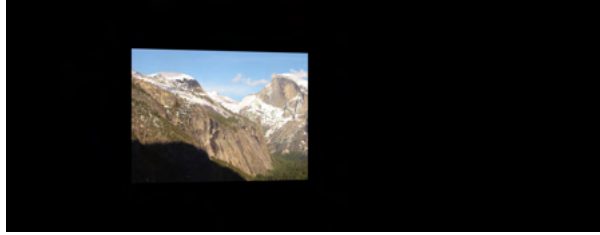
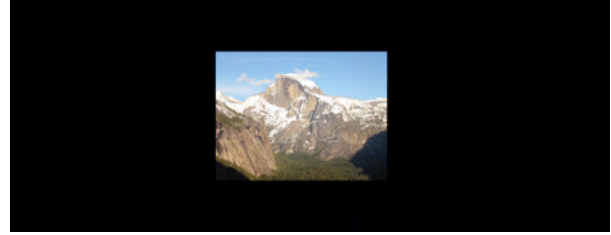


Figure 3: Feature matching results between input image 1 and 2.

Since these photos are taken at different angles, to be able to blend the images together we have to warp one image onto the plane of the other. This change in image domain aligns the features that were matched between images and allows us to blend the images without significant errors in the blended image.



(a) Image 1 warped onto the perspective of image 2



(b) Image 2 remaining on its perspective

Figure 4: Image warping results between input image 1 and 2.

Step 3: Image blending

Once the features of consecutive images are aligned through warping one image onto the perspective of another, they must be combined into a seamless output. Blending through adaptive weights is used, with the given function in the assignment, to provide a blend without visible seams between images. The input images to blend are the warped images shown above shown in Figure 4.

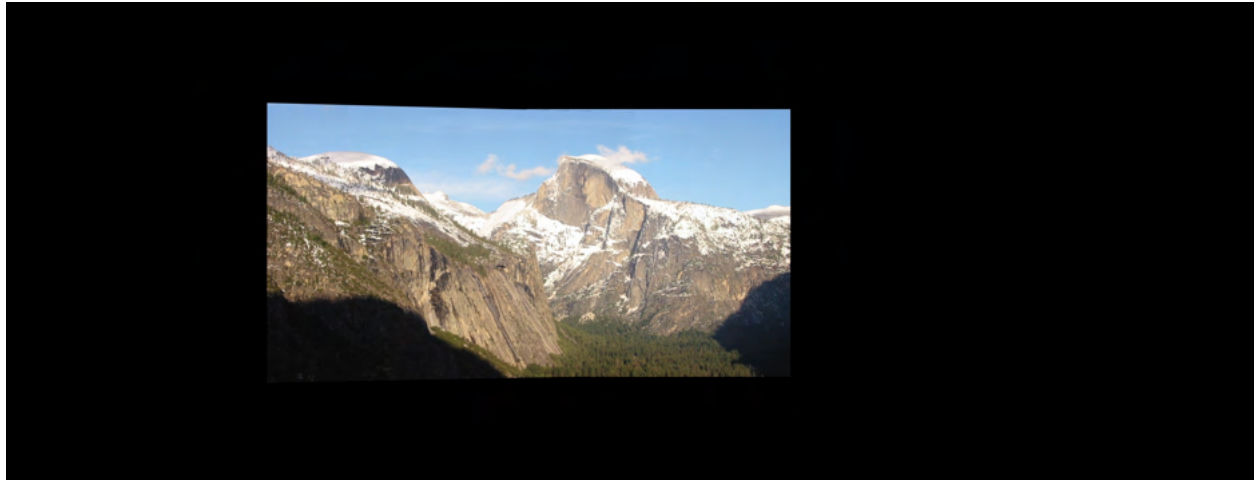


Figure 5: Distance transform blended result between input of warped image 1 and 2.

Step 4: Stitching multiple images

Stitching multiple images together creates one final panorama that includes all of the input images. To limit drift, we use pairwise alignment and matrix superposition on a chosen anchor image, chosen as image 2 here. Drift is limited, as the stitching is split into two halves of to the left and to the right of the anchor and so drift between image transitions accumulates less than if images had been stitched together left to right consecutively.

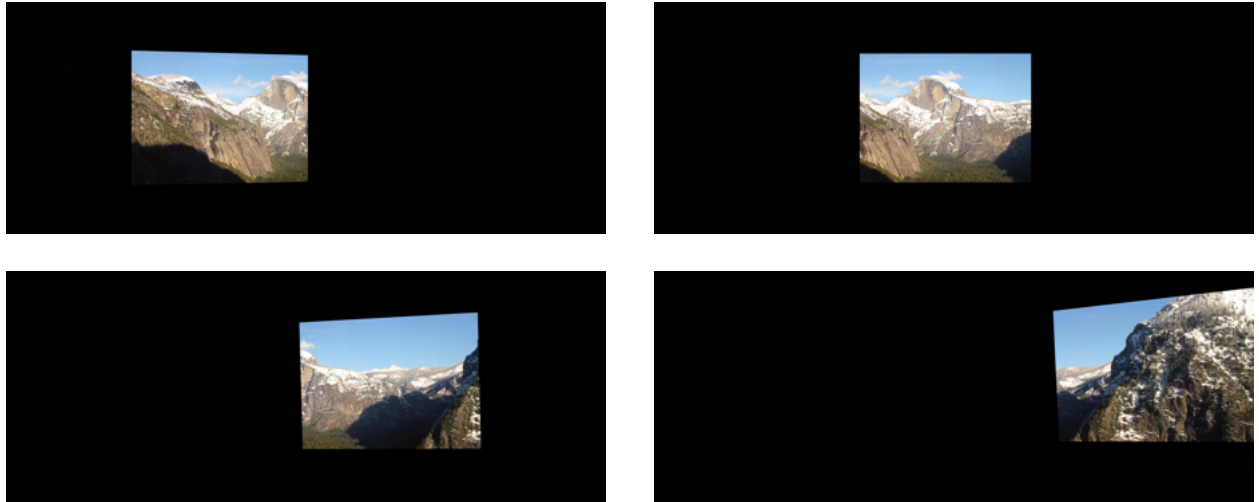


Figure 6: Input images to panorama stitching pipeline

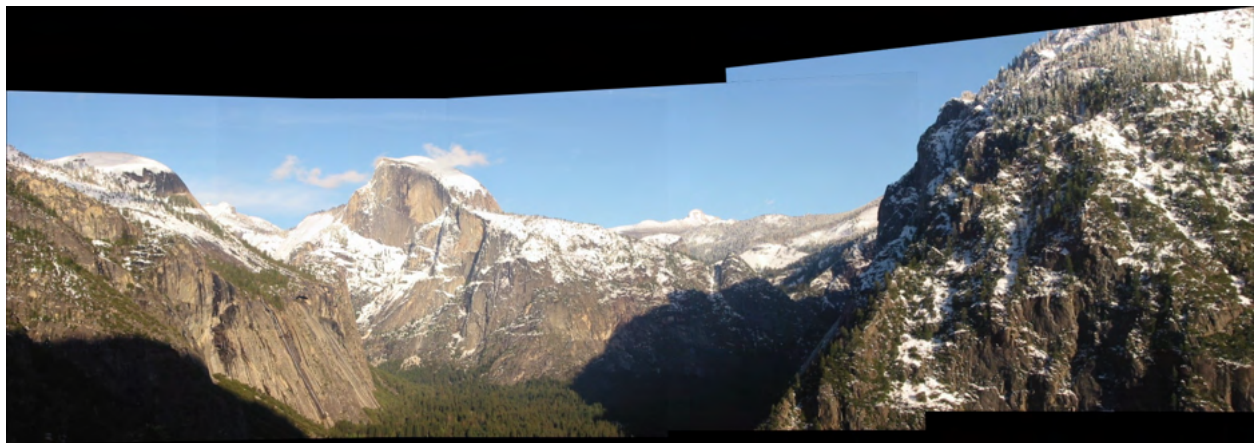


Figure 7: Result of stitching above images into panorama.

Step 5: Create your own panorama and discuss the setup and results



(a) Input image 1



(b) Input image 2



(c) Input image 3



(d) Input image 4

Figure 8: Input images to panorama stitching pipeline



Figure 9: Final panorama under original pipeline



Figure 10: Panorama with errors marked

Why each error occurs:

1. For the central error with the bike and the walkway, where the foreground is significantly blurred: The matching between images 2 and 3 from the original pipeline shows that matches are concentrated far away from the bike and the walkway and instead at the tunnel background and mural. This is unlike the matching between consecutive images of Yosemite, where matches were spread out generally evenly in the areas of overlap between images. Both the mural and background have repeating objects which caused matching errors between images, and so the small errors in alignment are seen through the bike and walkway appearing as blurry in the final panorama.



Figure 11: Image matching between images 2 and 3

2. For the error on the far right of the panorama, where the wall of the mural appears significantly longer than it actually is: Between input images 2 and 4 the camera was rotated significantly. When warping perspectives of the images, image 2 was used as the anchor, and so the rotation between photos caused the warped image to be rotated much more than image 1 or 3. When stitching all of the images together, the rotation in the warped image 4 stretches the plane and makes it look like we are looking at the wall from the side rather than the straight on angle of the rest of the picture, causing it to look longer than the wall really is.

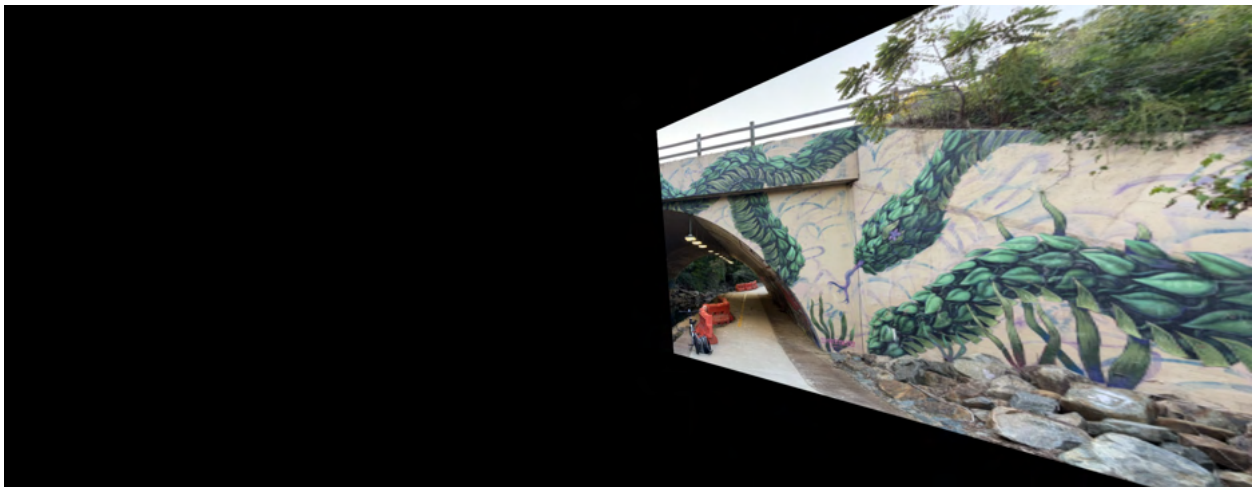


Figure 12: Image 4 warped onto the perspective of image 2

Resolving each issue:

Errors 1 and 2 match up with the errors described above.

1. A stricter RANSAC reprojection threshold, from 5.0 in the original pipeline to 1.0 in the new pipeline, is implemented. This change makes the feature matching more strict on what it considers a good match. Although the matches are still concentrated away from the bike and walkway, by aligning it only on matches that vary by 1 pixel instead of 5 the alignment of the images has less error and so the bike and

walkway is clear unlike in the panorama from the original pipeline.



Figure 13: Panorama with adjusted RANSAC threshold

2. The capture setup is changed to have less rotation between images. Despite the anchor image remaining as image 2, this change causes the warped image 4 to be rotated significantly less than from the original photos. From this, when all of the images are stitched together into the final panorama the wall mural on the right side appears at a normal length compared to how it actually looks. This change in photo capturing approach produced a significantly better final panorama, also removing the issue of blurriness near the bike and walkway.



Figure 14: Panorama with adjusted capture setup